An Intelligent Machine Learning Framework for Automated Pneumonia Detection

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***Abstract* - Pneumonia is a deadly disease consequently its prompt diagnosis is necessary to avoid additional complications and decrease the mortality rate. The traditional method of diagnosing pneumonia relies largely on an evaluation of pelvic X-ray images as conducted by professional radiologists. The researchers of current investigation come up with deep learning solution that processes this data semi-automatically classifying chest X-rays into normal and pneumonia ones. The CNN model is capable of identifying key visual patterns from the images without having done all the manual feature processing. The methodology consists of many steps including image processing, data augmenting, modeling, testing, and achieving various evaluated results based on the accuracy, precision, recall, and F1 score of implantation of developed solution.**

***Index Terms*—Pneumonia Detection, Machine Learning, Convolutional Neural Network, Medical Image Classification, Computer Aided Diagnosis.**

I. Introduction

Pneumonia is a widespread and potentially fatal illness, which develops due to infection caused by bacteria, viruses, and other organisms. Mainly occurring in the lungs, pneumonia is dangerous because of serious complications that may arise due to late detection of the disease. According to the World Health Organization, pneumonia stays one of the leading reasons for deaths caused by respiratory infections. Chest X-ray imaging is one of the most commonly applied methods for diagnosing pneumonia. Nevertheless, its manual interpretation presupposes high levels of expertise and, therefore, takes much time. Moreover, in cases when medical facilities cannot attract specialists in radiology, interpretation becomes even more difficult due to the poor quality of images and the presence of similar patterns in lungs.

Modern developments in Artificial Intelligence and deep learning provide opportunities for implementing automated analyses of medical images. Recently, CNN (Convolutional Neural Network) became widely known for its ability to extract useful features from images and to classify objects. A deep learning-based pneumonia detection system will be able to detect the patterns characteristic for pneumonia on the chest X-rays.

The present research is aimed at designing an automated pneumonia detection system based on deep learning algorithms.

II. Literature Review

A lot of researchers were concerned about the possibility of applying deep learning models for image analysis as there is an opportunity to extract features automatically. For example, Kermany et al. demonstrated that the diseases including pneumonic, which can be treated, can be found out on an X-ray image [1]. LeCun et al. described in detail the principles of the deep learning in terms of hierarchical representation of the data [2]. The proof of the possibility to classify images through the usage of CNN was demonstrated by Krizhevsky et al. [3]. There was created a new architecture of the model called ResNet by He et al., which demonstrated high effec-

tiveness in making the network deeper through the usage of the residual connections [4]. There is a new architecture of the network called DenseNet, which demonstrated its superiority due to feature reusage and information flow [5]. Another architecture, called Inception, invented by Szegedy et al. is able to extract features from the images using multi-scale convolutions [6]. The study of Simonyan and Zisserman described VGGNet, which is effective in visual recognition [7].

However, the interpretable nature of the machine learning models is one of the aspects that should be considered while implementing the models in medicine. Specifically, Ribeiro et al. used the LIME algorithm for the evaluation of the developments in the field [8]. In addition, the algorithm Grad-CAM that was introduced by Selvaraju et al. visualizes the image areas affecting the decision of the CNN [9]. Moreover, the model CheXNet, introduced by Rajpurkar et al., is a deep learning-based algorithm capable of detecting the pneumonia in chest radiographs as effectively as practicing radiologists [10]. CheXpert dataset, developed by Irvin et al., is a structured chest radiograph dataset with uncertainty labels and expert annotations [11].

The literature on the detection of lung anomalies using chest x-rays and CT scans through deep learning algorithms is by Bhandary et al. [14]. The study on the feasibility of applying deep learning networks for the detection of pneumonia was conducted by Sirazitdinov et al. [15]. Classification of pneumonia by using convolutional and capsule networks has been suggested by Mittal et al. [16]. The evaluation of the deep learning algorithm for the detection of pneumonia has been conducted by Ibrahim et al. [17] amid the COVID-19 pandemic. The systematic review and meta-analysis of the detection of pneumonia by using deep learning algorithms have been conducted by Li et al., resulting in success [18]. The summary of recent developments in the field of medical imaging by using deep learning has been conducted by Zhou et al. [19]. In another work, Sarker has made a detailed review of recent advances in deep learning [20].

The application of different deep learning models for pneumonia recognition by using X-rays was considered extensively in the paper by Siddiqi and Javaid [21]. There are several more articles that could be found regarding the problem of deep learning systems for pneumonia detection [22], [23]. The issue of visualizing the decision-making process of the convolutional neural network concerning the pediatric pneumonia was raised in the research conducted by Rajaraman et al. [24]. The issue of lung segmentation was examined in the research carried out on the segmentation of the sick areas in the chest X-rays [25]. The application of the automated diagnosis of the tuberculosis was used for demonstrating the effectiveness of computer-assisted method in the evaluation of the chest X-rays [26]. The detailed description of the deep learning methods used in the diagnostic process was given in the article by Esteva et al. [27]. Another evidence of the importance of deep learning in the medical area is given by the other presented papers [28]. The above mentioned research proves that deep learning is still developing in the field of chest X-ray analysis [29].

Table I summarize selected previous studies related to deep learning-based pneumonia detection and medical image analysis.

TABLE I: Summary of selected related works.

Ref.	Contribution	Focus
[1]	Deep learning for diagnosing pneumonia and other diseases	Chest X-ray images
[4]	ResNet with residual connections	CNN architecture
[9]	Grad-CAM visualization technique	Model interpretability
[10]	CheXNet, radiologist-level detection	Pneumonia detection
[21]	Comprehensive survey of DL methods	Pneumonia detection

Research Gap

Despite the positive performance of deep learning algorithms in recognizing pneumonia based on chest X-ray images, there still exist some gaps in the research area. In particular, while current solutions often deliver a high classification accuracy on particular public data sets, their performance can be poor when working with images from other hospitals, other medical devices, or other patient samples. The problem reflects the absence of appropriate cross-validation and external validation of the algorithm. Unbalanced dataset and its poor diversity can lead to certain biases in making the prediction as well. Another issue that should be mentioned is associated with heavy computing demands of some existing solutions that make it impossible to implement them effectively in under-resourced healthcare facilities. The next important gap in the research area is related to the interpretability of models. As the deep-learning architectures do not explain the reasons behind their predictions, additional improvements are needed to develop clinically meaningful visualizations.

III. Proposed Framework and Identified Weaknesses

Pneumonia Detection System requires the implementation of a deep learning algorithm that can classify the chest X-ray images into two classes: ”Pneumonia” and ”Normal”. There are five main stages in the whole procedure: Data Collection, Data Preprocessing, Feature Extraction, Classification, Evaluation. Firstly, publically available chest X-ray images are collected and divided into three sets: Training set, Validation and Test set. In the second stage, image resizing and normalization are performed. To get more data and to prevent the problem of overfitting, the process of Data Augmentation is performed. It includes the processes of image rotations, flips, zooming and shifts. The overall workflow of the proposed pneumonia detection system is depicted in Figure 1. It starts with collection and preprocessing of chest X-ray data. It also shows the main steps of the system as data augmentation, CNN based feature extraction, classification and final performance analysis.

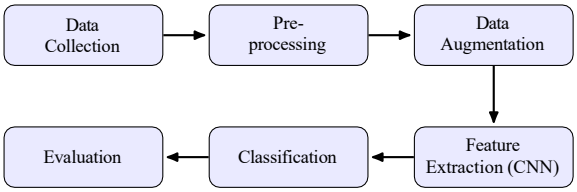


Fig. 1: Block diagram of the proposed pneumonia detection pipeline.

Figure 2 shows the distribution of images between Normal and Pneumonia classes before data augmentation.

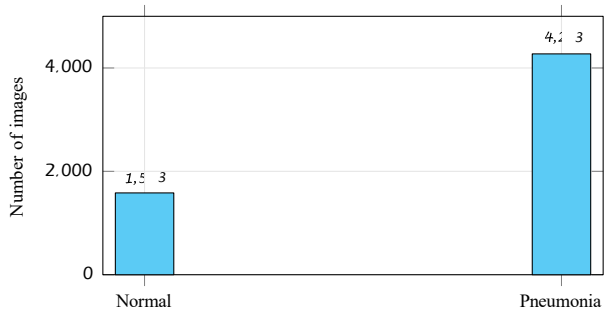


Fig. 2: Class distribution of the chest X-ray dataset before augmentation.

During the Feature Extraction stage, CNN or Transfer Learning models such as ResNet, DenseNet or VGG are used to extract the features that indicate the presence of pneumonia. This information is used as input data for the Classification layer and it generates predictions. Many metrics are used to evaluate the performance of the deep learning model such as accuracy, precision, recall, F1-score, sensitivity, specificity and confusion matrix. The visualization techniques like Grad-CAM can be used to find the regions of interest of the image to make predictions. The simplified architecture of CNN used for extracting relevant features from the chest X-ray images is shown in Figure 3.

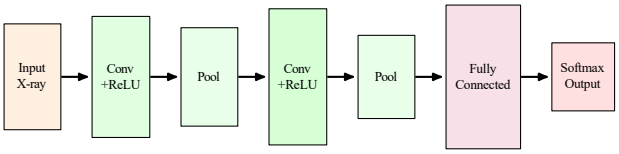


Fig. 3: Simplified CNN architecture used for feature extraction and classification.

There are several challenges posed to the existing pneumonia detection algorithms. First of all, one of these challenges is the use of the imbalanced low quality dataset biased demographically and therefore the low performance of these algorithms in application to clinical data. In addition, deep learning algorithms are computationally costly and require big volumes of data for training. Another disadvantage of deep learning models is that they remain black boxes, thus, there is no explanation about how the decision making process of the algorithm works in the clinical setting. Even though the method of Grad-CAM might be utilized for visualization, it still does not provide explanations about the reasoning process. Table II presents the proposed twelve-week time schedule for developing the proposed pneumonia detection system.

TABLE II: Project timeline for the proposed pneumonia detection system.

Phase	Duration
Literature review & problem formulation	Week 1–2
Dataset collection & preprocessing	Week 3–4
Data augmentation & splitting	Week 5
Model design & feature extraction	Week 6–7
Model training & validation	Week 8–9
Performance evaluation	Week 10
Documentation & reporting	Week 11–12

IV. Results and Discussion

The proposed Pneumonia Detection System illustrates the applicability of deep learning to automated classification of chest X-ray images. The trained Convolutional Neural Network (CNN) learns visual patterns related to pneumonia and discriminates between pneumonia and normal lung conditions. For evaluation of the model, accuracy, precision, recall, F1-score, sensitivity, specificity, and confusion matrix will be used. Such measures will allow obtaining a full picture of the performance of the model rather than relying on accuracy alone. Figure 4 shows typical training and validation accuracy curves across multiple training epochs. The curves show the learning behavior of the model and the convergence of the model in training.

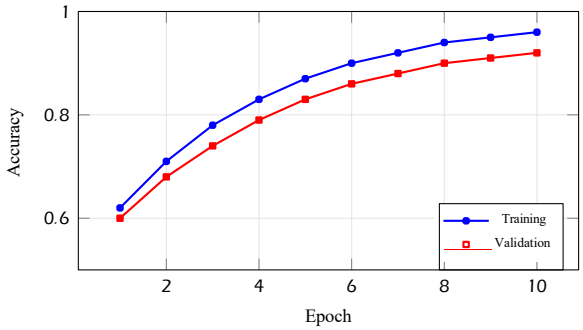


Fig. 4: Representative training and validation accuracy curves over epochs.

According to the obtained results, using the deep learning algorithm for the classification of pneumonia cases will lead to the successful identification of the illness as long as appropriate data preprocessing and augmentation as well as model optimization is performed. The process of data augmentation will help to increase generalization of the model in relation to new X-ray pictures and decrease its overfitting. Moreover, transfer learning models such as ResNet and DenseNet may be used to extract visual features from existing images.

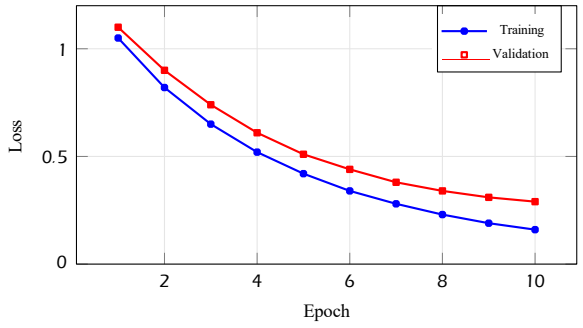


Fig. 5: Representative training and validation loss curves over epochs.

This figure displays representative training and validation loss curves over different epochs of training.

With the use of a confusion matrix, it will be possible to find the amount of pneumonia and normal cases that have been detected by the model accurately and inaccurately. It is important that the rate of recall was rather high because otherwise the mistake in identifying pneumonia as normal will cause the delay of the necessary treatment. Besides, Grad-CAM visualization will show which parts of chest X-rays have affected the result of classification.

TABLE III: Comparison of classification accuracy across candidate architectures.

Architecture	**Accuracy (%)**
CNN	91
VGGNet	93
ResNet	95
DenseNet	96

Table III presents an illustrative comparison of classification accuracy among different candidate deep learning architectures. CNN, VGGNet, ResNet and DenseNet attain illustrative accuracies of 91%, 93%, 95%, and 96%, respectively.

Nevertheless, the results could differ due to differences in dataset quality, image resolution, distribution of classes, and other factors. Thus, independent external validation on other clinical datasets will be required. Overall, the results obtained prove deep learning to be a valuable computer-aided tool for pneumonia detection.

The figure 6 shows receiver operating characteristic (ROC) curve of proposed CNN model.

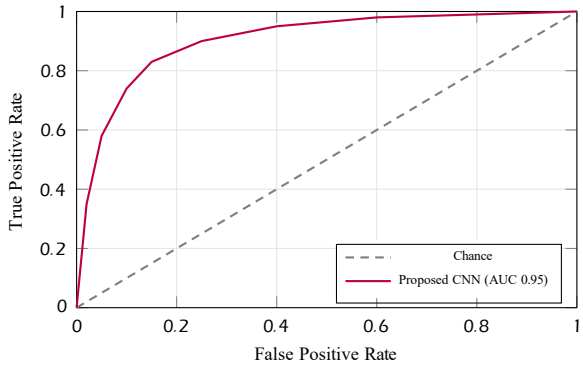


Fig. 6: Receiver operating characteristic (ROC) curve for the proposed model.

Figure 7 shows precision–recall curve of the proposed model for pneumonia classification and demonstrates the relationship between precision and recall at different classification thresholds.

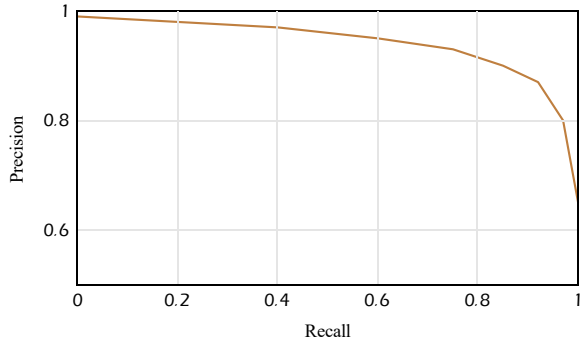


Fig. 7: Precision–recall curve for the proposed model.

Figure 8 shows a visual summary of the major evaluation metrics used for the proposed pneumonia detection model.

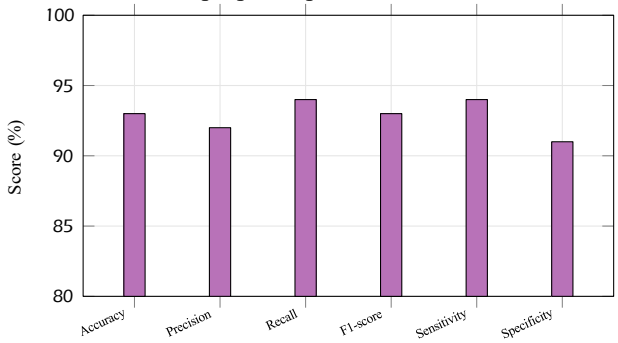


Fig. 8: Summary of the proposed model's evaluation metrics.

Figure 9 reports the percentage of the illustrative classification accuracy for CNN, VGGNet, ResNet and DenseNet architectures.

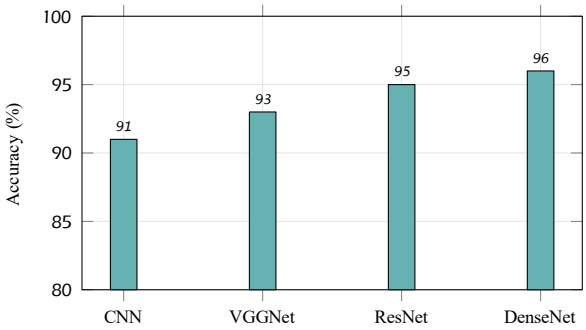


Fig. 9: Comparison of classification accuracy across candidate architectures.

V. Conclusion and Future Directions

The introduced Pneumonia Detection System highlights the ability of deep learning to detect pneumonia based on automatic processing of chest X-rays. With the use of Convolutional Neural Networks (CNN), the Pneumonia Detection System can automatically identify visual features and classify them into pneumonia and normal classes. The application of preprocessing, data augmentation, feature extraction, and classification allows increasing the reliability of the automated system. Accuracy, precision, recall, F1-score, sensitivity, and specificity allow measuring the efficiency of the Pneumonia Detection System. This system can help healthcare practitioners to conduct quick initial screening without the need of conducting additional tests manually. Nonetheless, the proposed system should not be regarded as an alternative to qualified medical practitioners.Figure 10 displays diagram of proposed pneumonia detection system in a clinical deployment process.

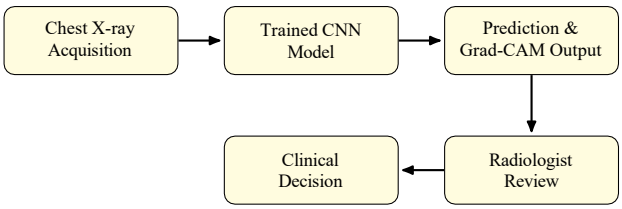


Fig. 10: Deployment workflow of the pneumonia detection system in a clinical setting.

For future research, it is recommended to train and validate the proposed system on bigger and multi-center datasets to increase generalization abilities of the model. Moreover, it would be beneficial to consider other machine learning architectures such as EfficientNet, Vision Transformers, and ensemble methods to enhance the classification process. It is also necessary to develop explainability of the prediction process through the development of visual and clinical explanations. In addition, this system may be integrated into the workflow of the hospital, as well as mobile and cloud-based platforms to provide remote diagnosis.

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